

Ilesh Dhall

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EDUCATION

B.Tech in Computer Science & Engineering, Graphic Era (Deemed to be University) July 2023 – Present
Specialization in Artificial Intelligence & Machine Learning **CGPA: 8.85 / 10.00**
Secondary & Senior Secondary, Bawa Lalvani Public School (CBSE) 2021, 2023
Science (PCM) with Information Technology **Class X: 95.17% | Class XII: 82.5%**

EXPERIENCE

Research Intern June 2026 – July 2026
Machine Learning Research Group (MLRG), Graphic Era (Deemed to be University)

- Researched **efficient LLM inference** with a focus on **KV cache compression** and long-context optimization.
- Integrated **exact attention saliency computation into FlashAttention** for memory-efficient long-context inference.
- Evaluated **GPU memory, latency, context length, and model accuracy** across multiple LLM benchmarks.

PROJECTS

AgriVerse – Multi-Agent AI Agricultural Platform April – May 2026

- Architected a **multi-agent AI system** with **4 specialized agents** orchestrated through workflow automation.
- Built a **FastAPI** backend and **Vite** interface supporting real-time interaction with multiple AI agents.
- Implemented a **RAG pipeline** using transformer-based embeddings and ChromaDB vector search.
- Containerized and deployed services on **GCP** using a microservice-based architecture for scalable serving.

Story-Forge – Fine-Tuned LLM Story Generation Platform May 2026

- Fine-tuned **Microsoft Phi-3 Mini (3.8B)** using **QLoRA** with **4-bit NF4 quantization** for memory efficient training.
- Applied **LoRA adapters** on transformer projection layers and merged trained adapters into the base model.
- Optimized inference using **Q4_K_M GGUF + llama.cpp**, enabling lightweight CPU-based deployment.

Deep Neural Compression for Neural Style Transfer October – December 2024

- Performed **layer-wise sensitivity analysis** across **16 convolutional layers** and evaluated **96 pruning configurations**.
- Applied **magnitude pruning** guided by sensitivity results on a **VGG19 model initialized with Caffe weights**.
- Achieved an **85% reduction in model size** and **3.2× memory reduction** while preserving stylization quality.

Metabolomics Data Analysis for Biomarker Discovery April – May 2025

- Processed a metabolomics dataset containing **23,000+ samples** across **80+ metabolite features**.
- Built a preprocessing pipeline with **feature pivoting, outlier capping, correlation filtering, and SMOTE balancing**.
- Trained and ensembled **Logistic Regression** and **Random Forest** with **cross-validation**, achieving **89% F1-score**.
- Deployed an interactive analysis dashboard using **R Shiny** for visualization and biomarker exploration.

PUBLICATIONS

SkyRis: An AI-Powered IoT Framework for Real-Time Cloudburst Prediction and Alerting 2026 |  DOI
Published in **IEEE ICCSC-2026 (Scopus Indexed)**; developed an AI-IoT cloudburst prediction system. **Best Paper Award**.

Exact Attention Saliency for Memory-Efficient Long-Context LLM Inference *Under Review*
Enabled **exact attention saliency estimation** within **FlashAttention** for memory-efficient long-context LLM inference.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++
ML & AI: PyTorch, Scikit-learn, Transformers, LLMs, RAG, Fine-Tuning, Quantization, Model Compression
Data & Analysis: NumPy, Pandas, OpenCV, Feature Engineering, Statistical Analysis
Backend & Systems: FastAPI, Docker, Linux, MySQL, Git, GCP

ACHIEVEMENTS

Amazon Machine Learning Summer School 2026
Selected among **3,000 participants** from **130,000+ applicants**, covering advanced topics in deep learning, GenAI, and LLMs.

2nd Place – alrIEEEena (State-Level ML Competition), GEHU Dehradun 2026
Ranked **2nd among 100+ participants** by building a high-performing machine learning model for multiclass classification.